Alec Zhou

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Education

University of Maryland -- College Park College Park, MD

B.S. Computer Science; B.S. Mathematics; GPA: 3.88 *Expected May 2027*

- **Coursework:** Algorithms, Operating Systems, Distributed Systems, Database Design, Compilers, Deep Learning

Experience

M&T Bank Buffalo, NY

Software Engineer Intern *June 2026 -- August 2026*

- Shipped a **production API-key service** using Spring Boot and PostgreSQL, implementing key generation, validation, rotation, and revocation with unit and integration coverage
- Implemented Redis caching of valid and invalid API keys for a production validation service handling **millions of requests per day**
- Delivered REST API and SQL changes across **15+ core-banking Java microservices**, integrating an external transaction-data provider with internal services
- Upgraded legacy services to Spring Boot 3.5, reducing Artifactory-reported dependency vulnerability findings **from 200+ to zero**
- Drove issue triage, implementation, and code review with OpenCode harness and GitLab Duo AI agents, validating agent-generated code before integration

Blake Willson Group Arlington, VA

AI and Process Automation Intern *June 2025 -- August 2025*

- Built a Python/LangChain RAG system with semantic retrieval and fuzzy search over internal contract data, adopted by all contract-facing staff to replace manual lookup
- Built a human-reviewed proposal-drafting pipeline that grounded drafts in retrieved contracts and aligned them with proposal requirements, saving an **estimated 4--5 hours per run**

Oceus Networks Herndon, VA

Software Engineer Intern *July 2023 -- August 2023*

- Automated bulk asset entry with Python/Pandas, replacing ∼45 minutes of manual entry per document with **uploads completing in under 10 seconds**; adopted by the warehouse team
- Built pages and dashboards for tracking different asset types in the company-wide PHP/Laravel/MySQL asset-management application

Projects

dbb -- Relational Database | *Rust* April -- July 2026

- Built a relational database engine in Rust with B+ tree indexing, SQL parsing, joins, Volcano execution, and query optimization through predicate pushdown and index selection
- Implemented buffer pool management with LRU page replacement and disk writeback; tested B+ tree integrity and end-to-end SQL execution

gatoway -- LLM Gateway | *Python, FastAPI, PostgreSQL, pgvector* July 2026 -- Present

- Built an OpenAI-compatible gateway routing across **8 LLMs** using embedding similarity and session-aware quality signals
- Cut **estimated inference cost 26.5%** with comparable mean quality in routing-policy benchmark screening
- Implemented pgvector-backed routing, provider failover, and automated outcome feedback; measured **34 ms p95 gateway overhead** in local benchmarks

gritter -- Codebase Q&A CLI | PyPI | *Python, tree-sitter, ChromaDB* April -- August 2026

- Published a codebase Q&A CLI to PyPI, combining AST-aware chunking, hybrid dense/BM25 retrieval, and verified file/line citations across multiple LLM providers

Technical Skills

Languages: Java, Python, Rust, SQL, TypeScript, C/C++
Frameworks & Libraries: Spring Boot, FastAPI, LangChain, PyTorch, Pandas, React, Next.js, Node.js
Databases: PostgreSQL, pgvector, MySQL, SQLite, Redis, MongoDB
Cloud & Tools: Azure, AWS, Docker, Git, CI/CD, Linux/Unix
AI & Agents: LLM agents & tool-use, MCP, RAG, GitHub Copilot, Cursor, Claude Code